

AI-POWERED TUTOR

TEAM - **BIGHERO4**



PROBLEM STATEMENT

Build a Smart AI tutoring app Built for Students that answers their questions, generate quizzes, and explain concepts and many more using generative text models running locally.



MOTIVATION

The primary motivation is to empower students and lifelong learners by leveraging the power of on-device Large Language Models (LLMs). By running AI **locally**, we can provide a **secure faster** and **responsive** educational tool.

100% Privacy & True Personalization:

All data and materials stay on the device, ensuring **privacy**, while AI references only what **user uploads** for context-aware assistance.

Offline Accessibility:

The app is fully functional without an internet connection, making learning possible in any environment.

High Performance:

Utilizing dedicated hardware like the **Neural Processing Unit (NPU)** via the Qualcomm AI Runtime (QART) ensures a smooth and efficient user experience.





Features Incorporated

- ◆ **01** Intelligently summarizes large documents to fit into AI model's context window.
- ◆ **02** Context-based instant summary, flashcards & quiz generation along with saved comprehensive review system of past – attempted quizzes.
- ◆ **03** Dual AI Modes & Real-time Chat: Global AI for cross-subject knowledge, & Subject-Specific AI for focused context.
- ◆ **04** Progress Tracking, and a Gamified User Profile with leveling, stats, and customizable goals.
- ◆ **05** Flags concepts needing re-practice based on quiz performance



Development Timeline

Phase-1

- Evaluated and finalized the core model (Llama-3.2) and libraries (PDF-Box) based on project requirements.
- Started working on a sample example repository to get familiar with the framework and environment

Phase-2

- Loaded the model through **Genie** and successfully ran it on the **NPU**.
- Based on it, we created a simple **chatbot** with context based on uploaded pdfs.

Phase-3

- Incorporated **pdf-context** based **Quizzes, Flashcards**, and enhanced the chat functionality.
- Implemented performance-driven **progress reports** and personalized **study goals**.

Phase-4

- **Intelligent summarization** to manage large document context effectively.
- Added a **concept-review module** that flags weak topics and enables focused, topic-specific chat.



A decorative abstract splash of colorful liquid, primarily in shades of blue, green, and yellow, is positioned on the left side of the slide, partially overlapping the 'Methodology' box.

Methodology

◆ 01

Model and Framework

◆ 02

Block Diagram

◆ 03

How to use

◆ 04

Parameters considered
and processing



Platform Android 
(Java & XML for UI/App Logic)

AI Framework Qualcomm AI 
Runtime (QAIRT)
for optimized on-device model execut

AI Engine: Genie SDK 
to manage model loading and inference.

Data Ingestion pdfbox-android 
for parsing and extracting text from user-uploaded PDF files.

Models & Framework

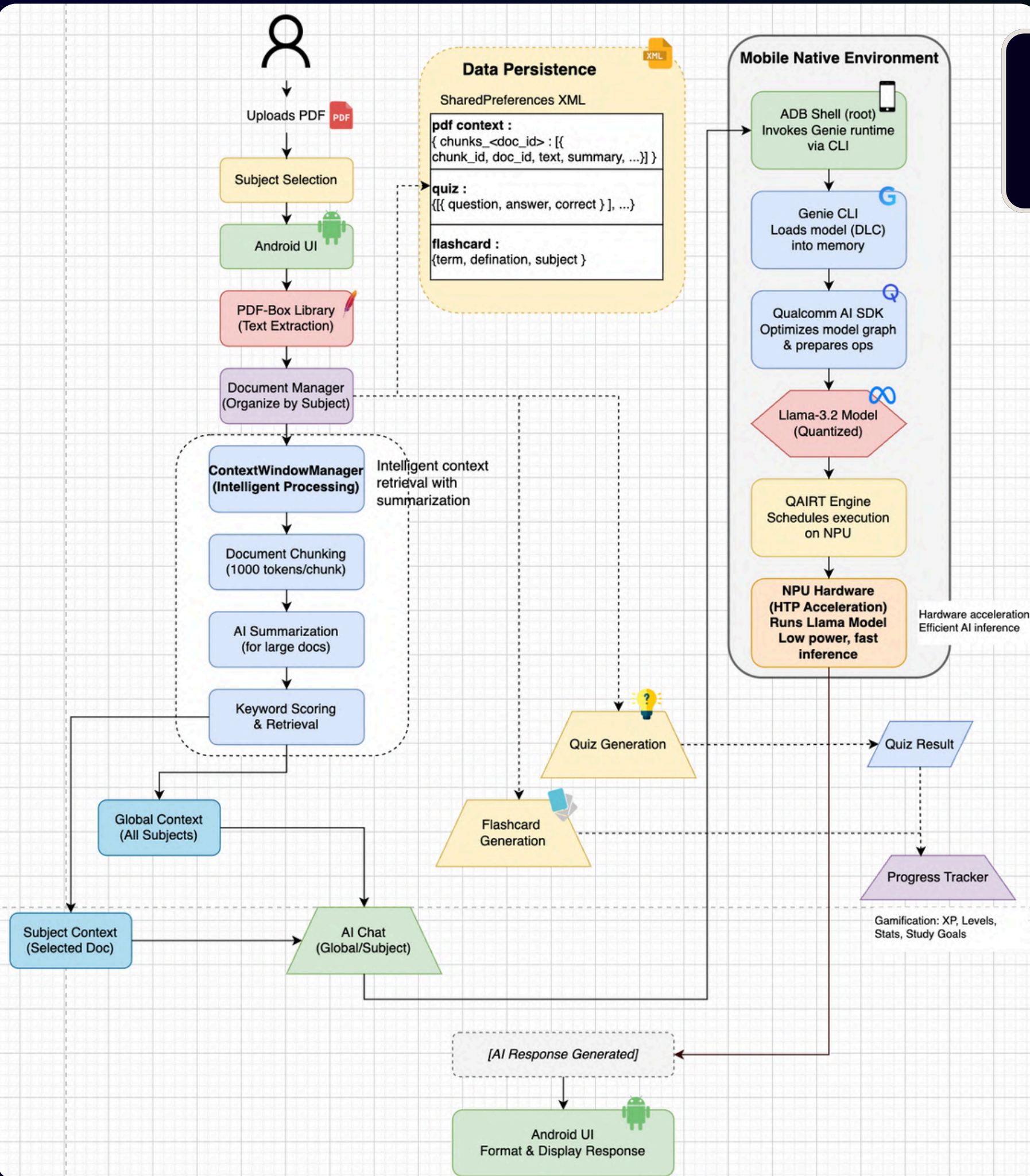
Core 
Model: Llama-3.2-3B instruct

Native Layer: C++ with JNI 
to bridge the Java application logic with the high-performance native AI engine.

Build System: Gradle with CMake 
for compiling native C++ code into a shared library (libchatapp.so).



Block Diagram



How to use the app

1. Set Up Profile First:

Tap the Profile icon. Add name, (optional : profile-photo, email, phone-no) and, most importantly, set learning preferences. This helps the AI give answers in the style one like best.

2. Add Study Materials :

Go to the "Subjects" tab. Create your subjects, then tap a subject to upload your PDFs. The app will automatically read them and prepare your study tools (in a few seconds).

3. Start Studying! One can study in two ways:

- Focused Study (One Subject):
 - Go inside a subject created.
 - Select "Generate Quiz," "View Flashcards," or "Ask Atom AI" to study only the documents in that single subject.
- Global Study (All Subjects):
 - From the Homepage, use the "AI Chat," "Interactive Quizzes," or "Learning Flashcards" buttons.
 - These tools will use content from all the PDFs you've uploaded, combined. We can review weak topics (where we performed poorly in quizzes).

4. Track Your Progress :

Tap the "Progress Overview" to see all stats, like study streak, past quiz scores, subject mastery. In "Profile" tab, one can see curated Study Goals.

5. Customize App :

Go to Settings to switch to Dark Mode for comfortable night studying.



Parameters Considered

- Relevance: Quality and contextual accuracy of the AI's answer.
- Model loading time : Time taken to fully load the AI model into memory before it can process requests.
- Time to First Token (TTFT): Time until the AI's response begins to appear.
- Inference Speed: Rate at which the model generates output, measured in tokens per second.

Preprocessing

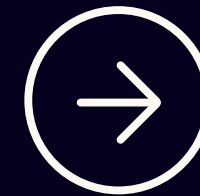
- PDF Parsing: Extracts text using pdfbox-android, organized by subject.
- Intelligent Chunking: Splits documents into 1000-token semantic chunks with AI-generated summaries for large texts.
- Context Indexing: Keyword-based scoring (10pts/match) identifies relevant chunks, cached in SharedPreferences for instant retrieval.
- Prompt Engineering: Combines queries with top-scoring context, formatted by PromptHandler.cpp for Llama-3.2 with anti-hallucination guardrails.

Postprocessing

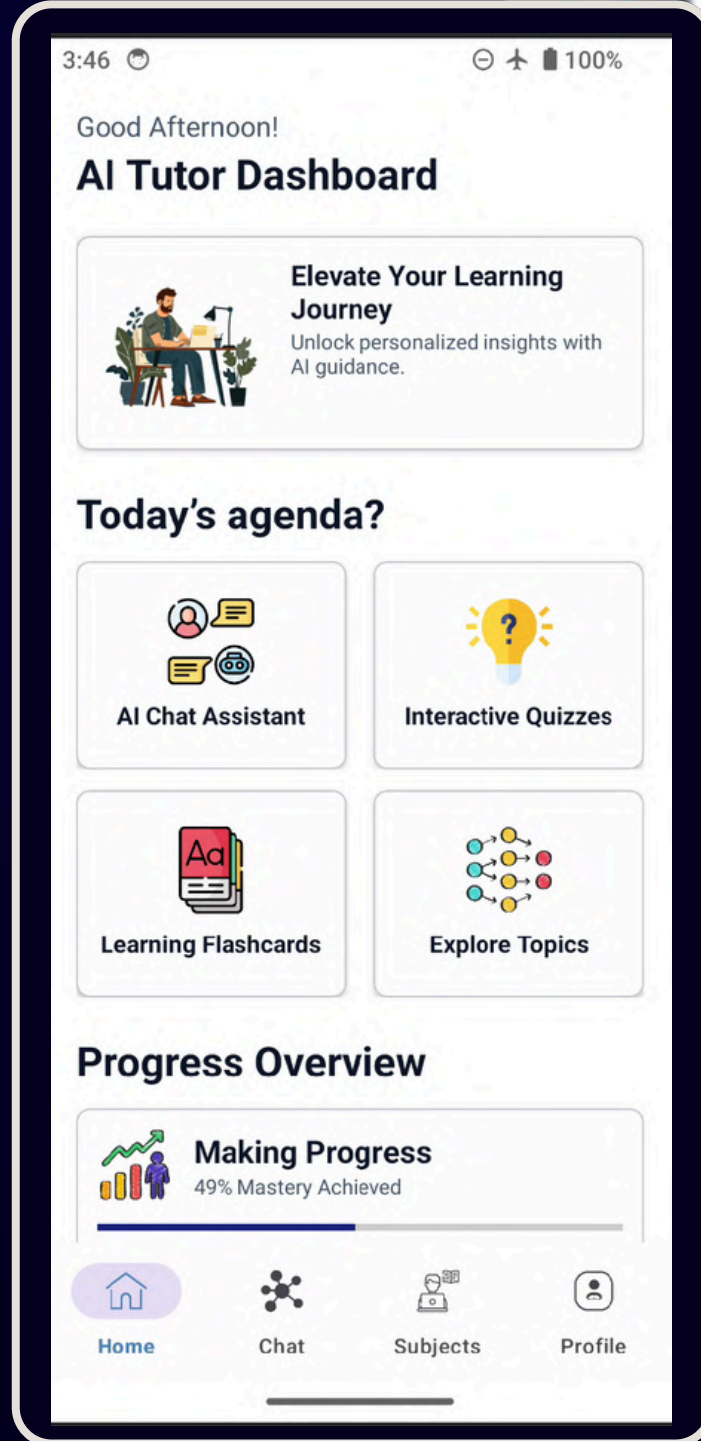
- Response Cleaning: Strips [END] markers, formats for UI display
- Smart Parsing: Quiz/Flashcard extraction → structured JSON storage
- Progress Tracking: Auto-logs scores/interactions → gamified metrics
- Context-Aware Routing: Global vs Subject-specific response handling



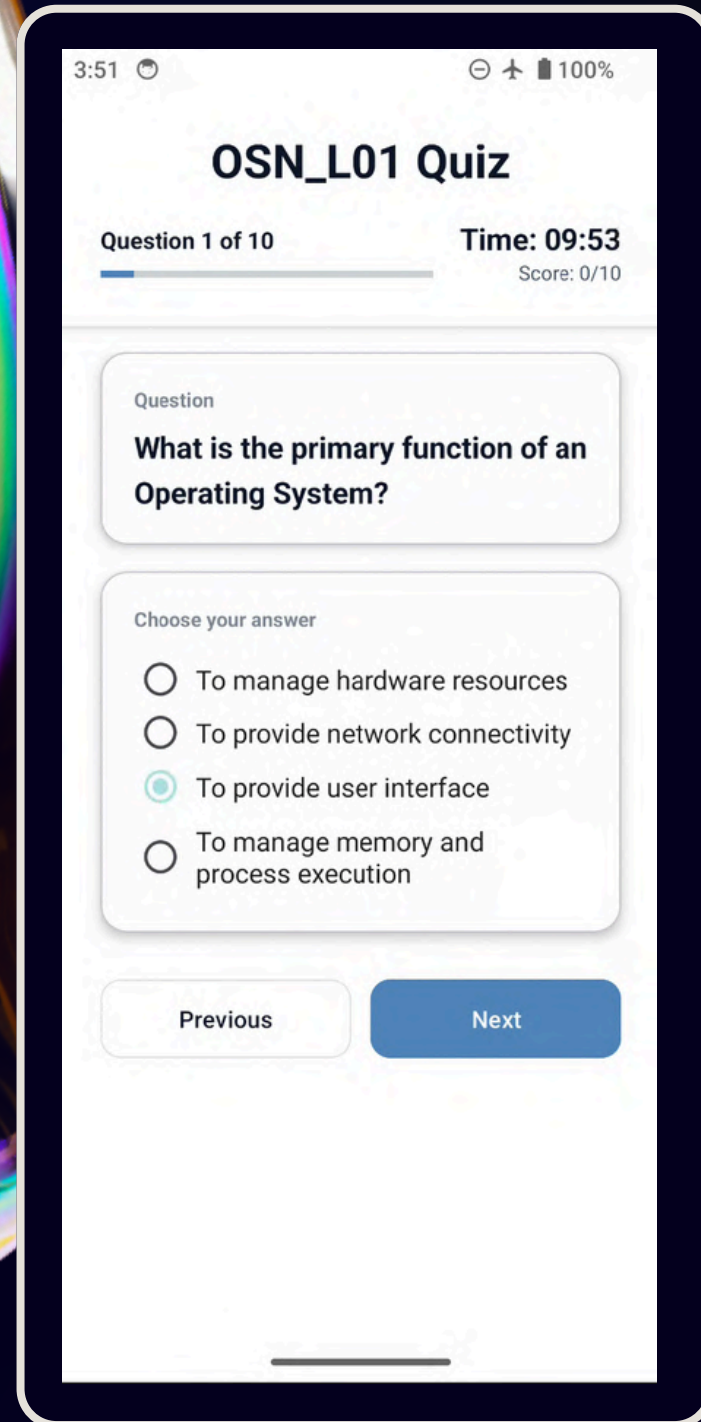
DEMO



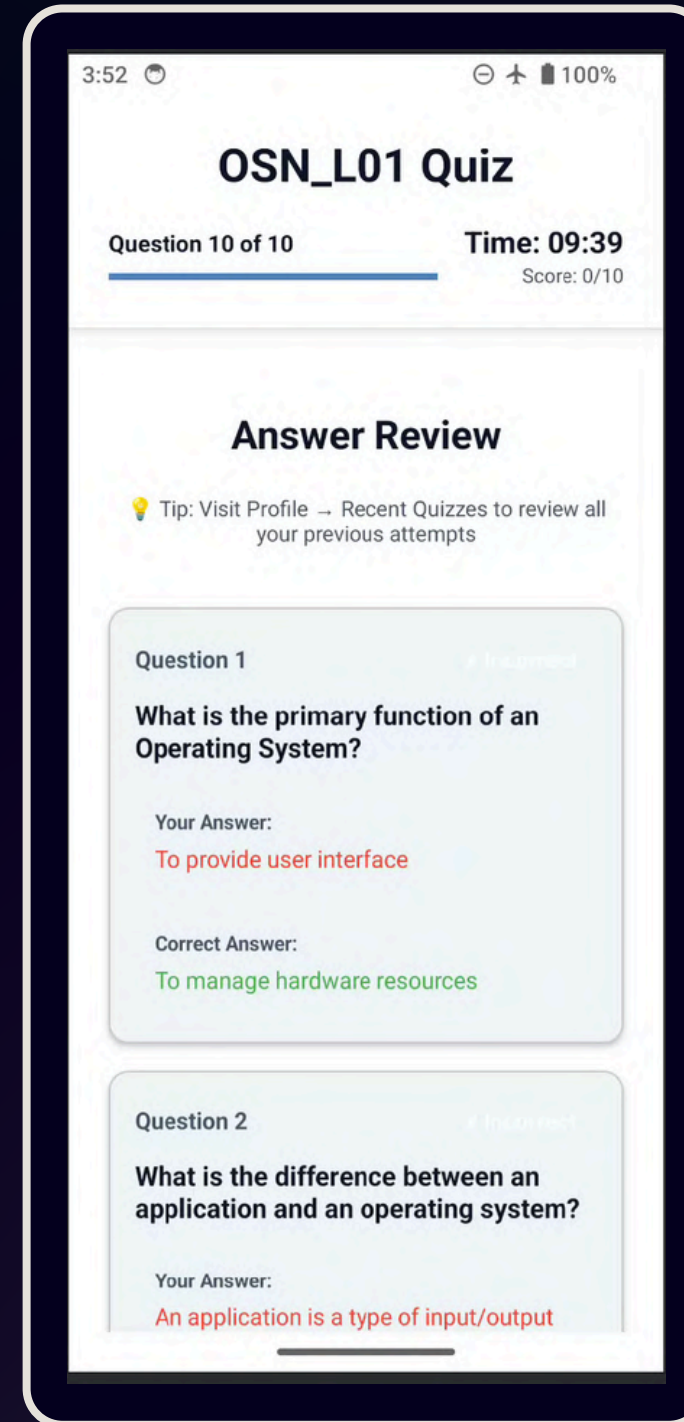
Homepage



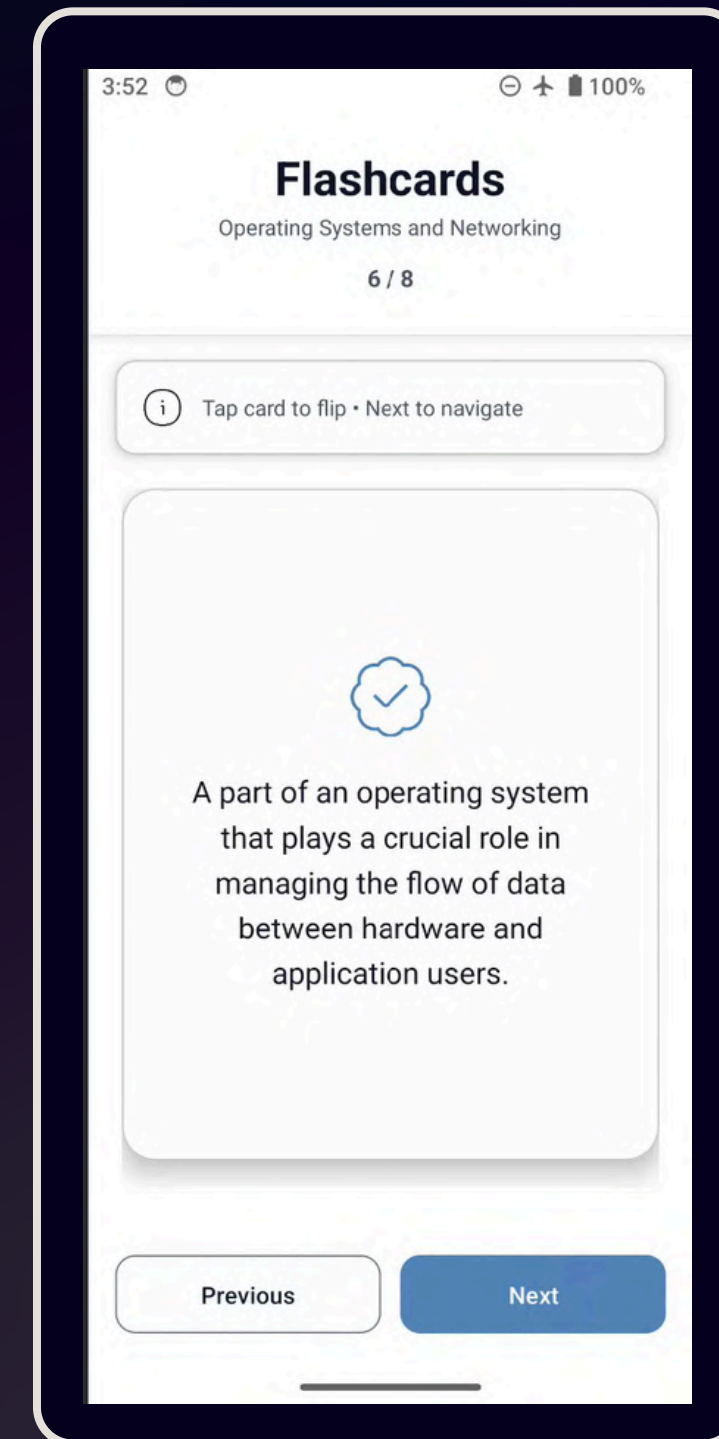
PDF-specific Quiz



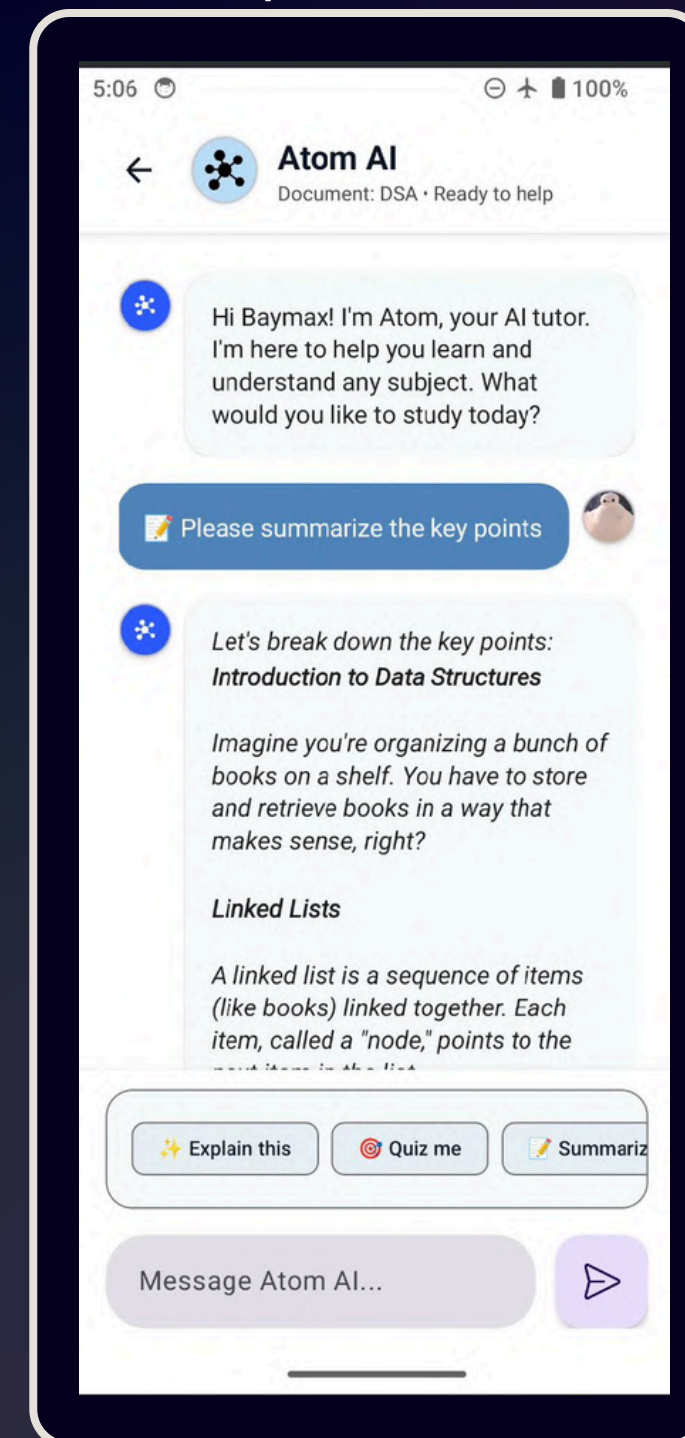
Quiz Review



PDF-specific Flashcard

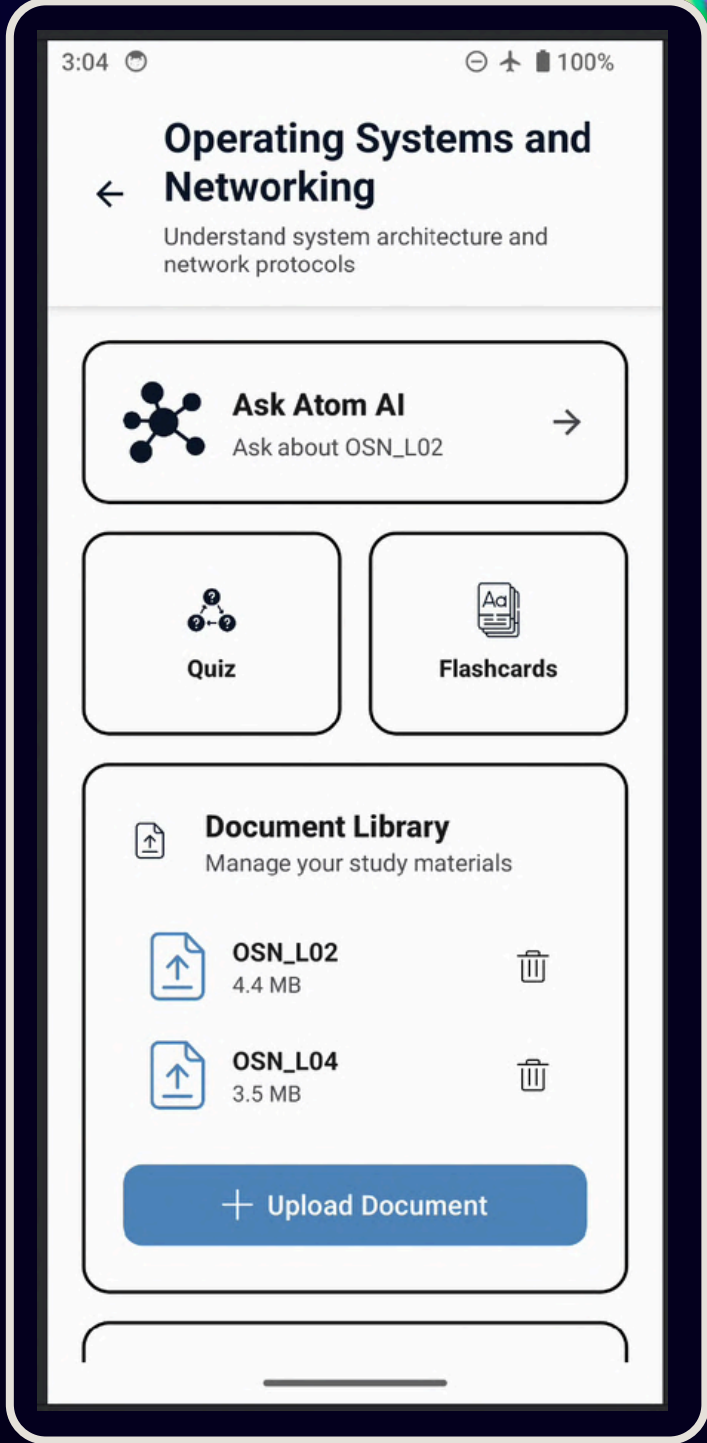


PDF-specific Chat

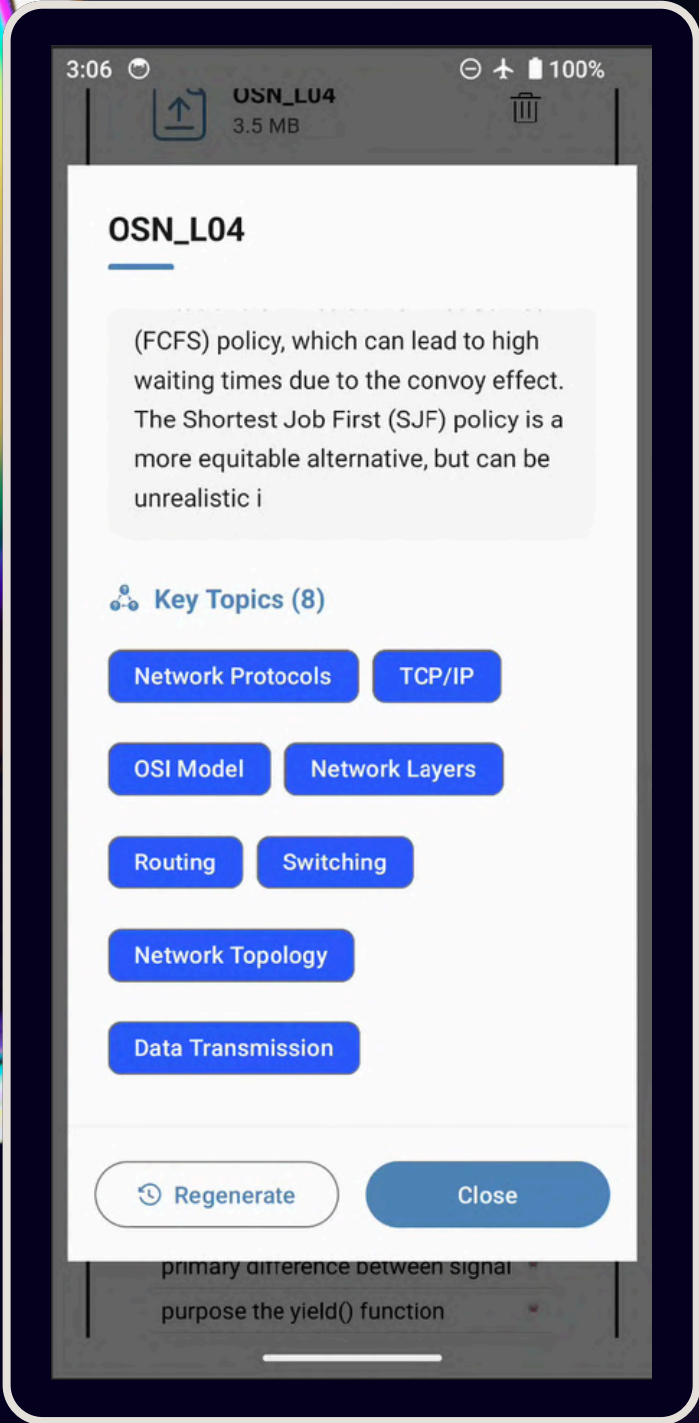


APP FUNCTIONALITIES

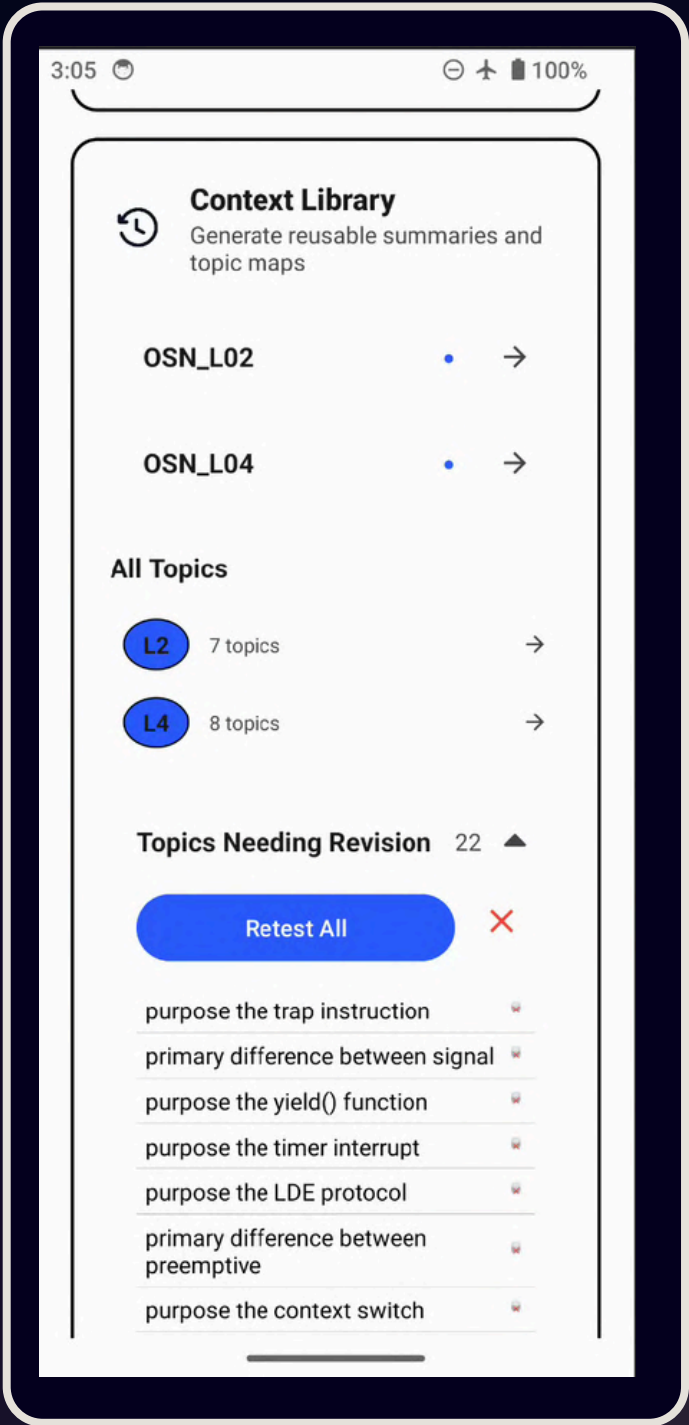
Subject interface



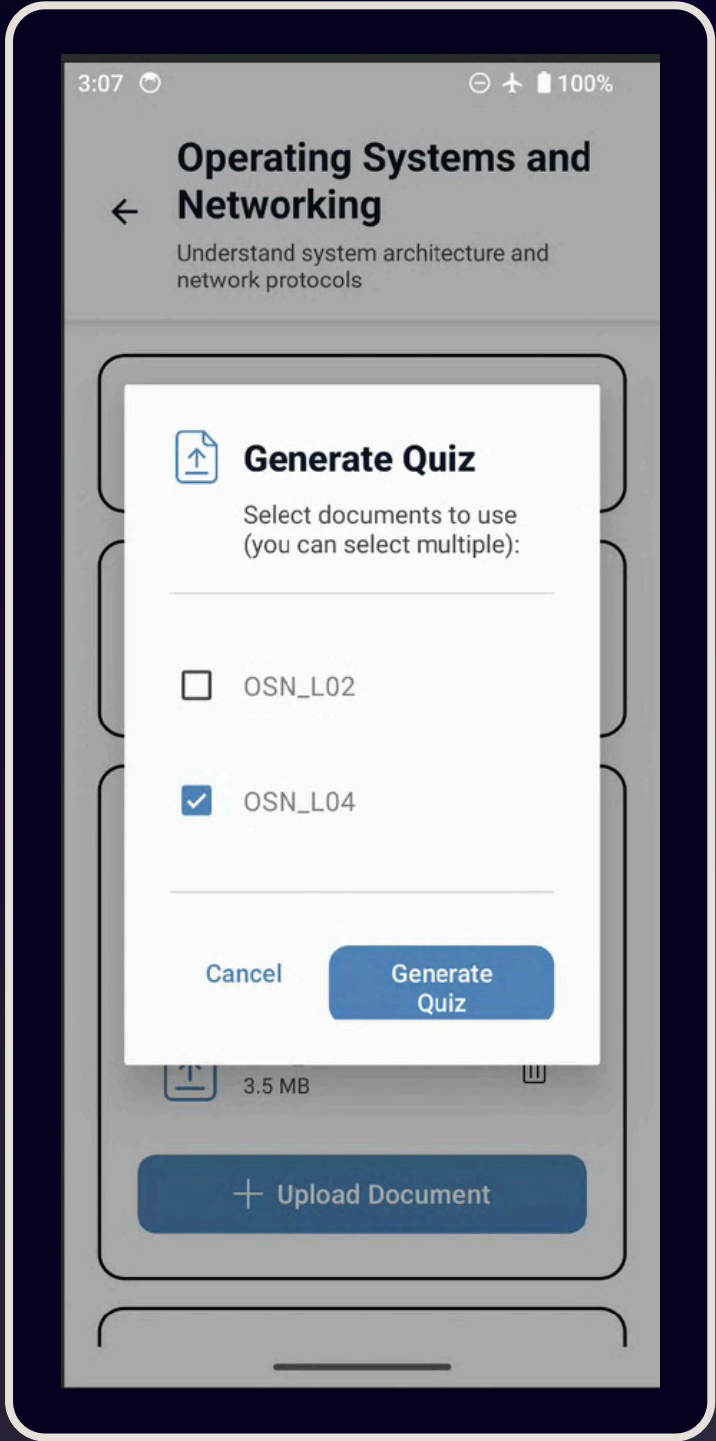
Quick Summary



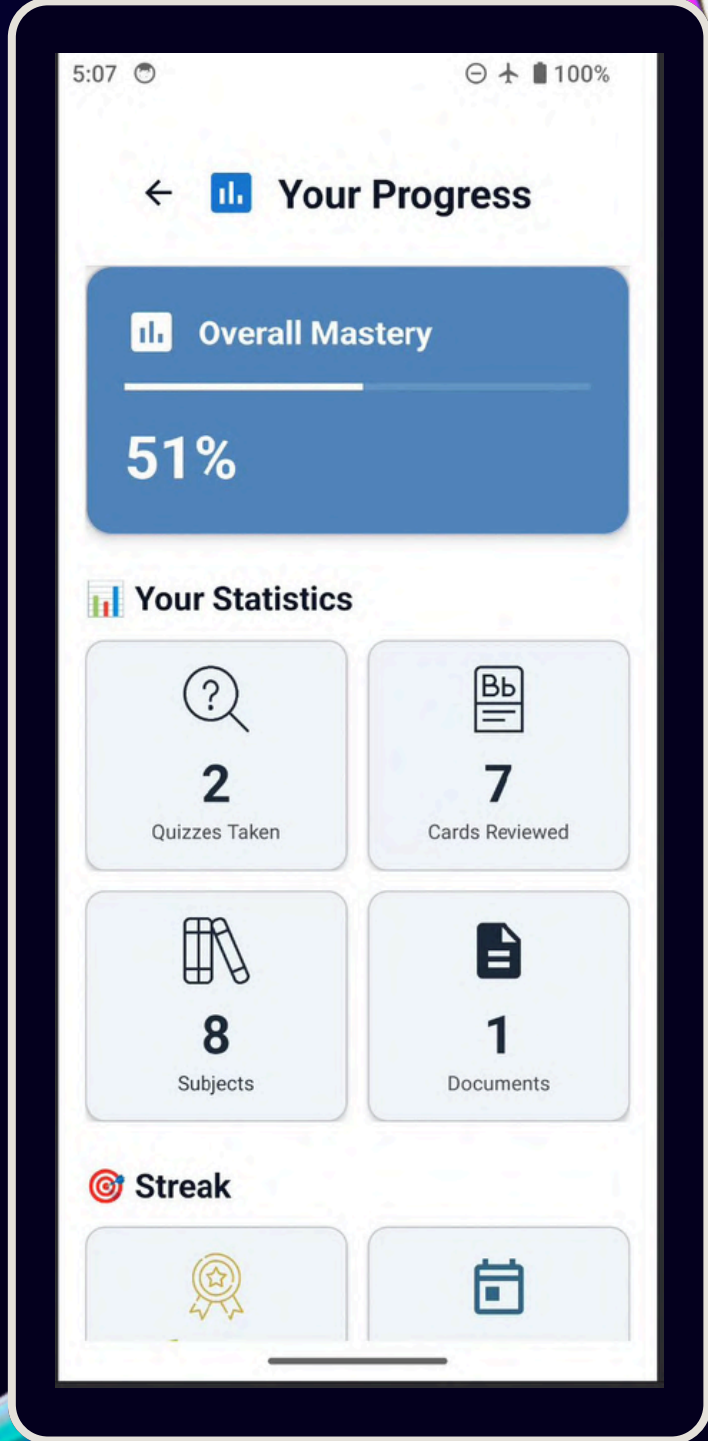
Weak Topics List



Fully personalized Quiz



Progress Report



Customizable Profile

5:06 100%

Basic Details
Update your personal details

Full Name
Baymax

Email Address (Optional)
bighero6@disney.in

Phone Number (Optional)
123-4567-890

Learning Preferences
Help AI personalize your experience

Preferred Difficulty Level

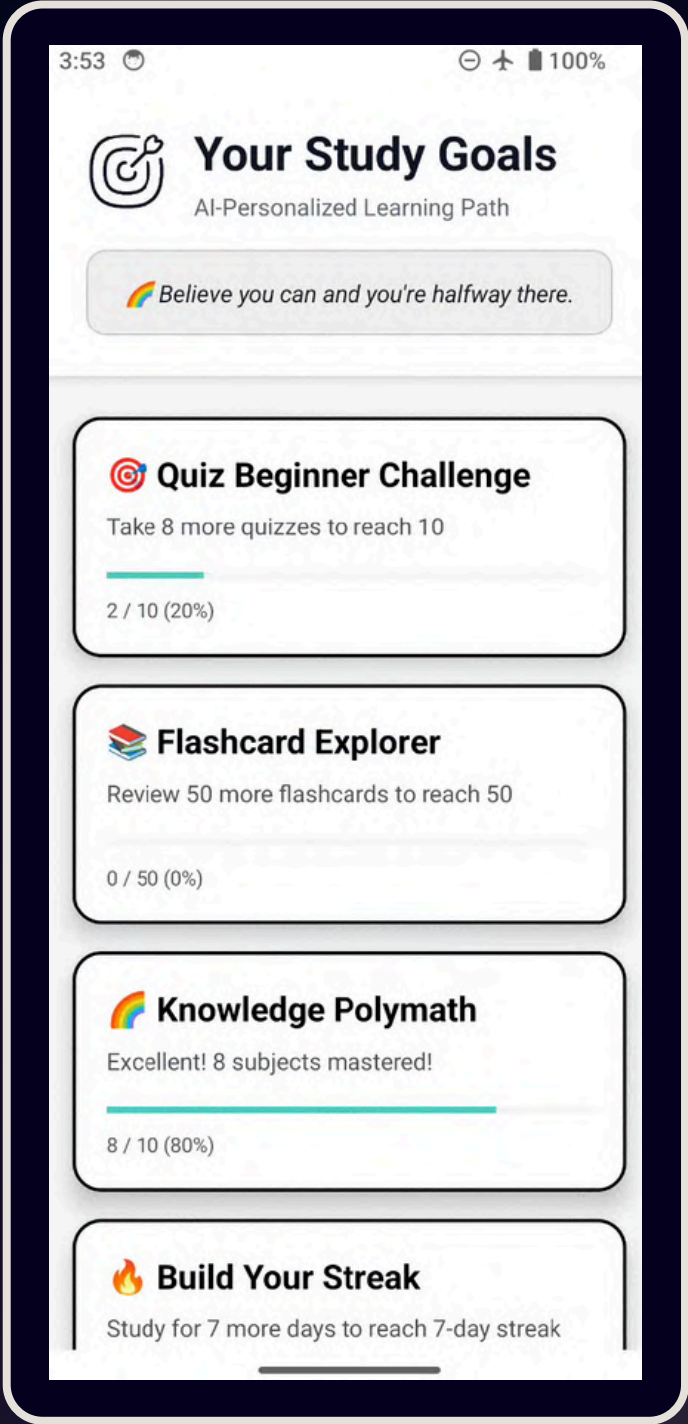
- ☐ Beginner - I'm just starting out
- ☐ Intermediate - I have some knowledge
- ☒ Advanced - I need in-depth concepts

What are you preparing for?
IIT College exams and Placement Interviews

e.g., College exams, Job interview, Certification, Personal learning

APP FUNCTIONALITIES

Study Goals



Profile Page

3:14 100%

Profile

Babymax
Learning Explorer • Level 4 • Building foundations ★

Personal Info
Update your name and email

Your Statistics

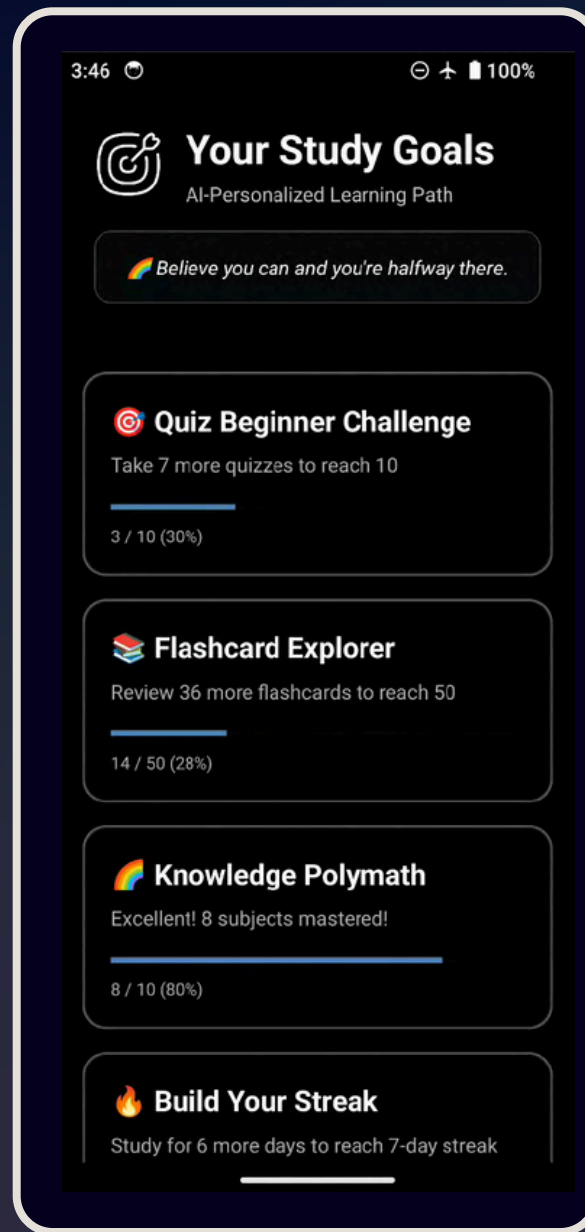
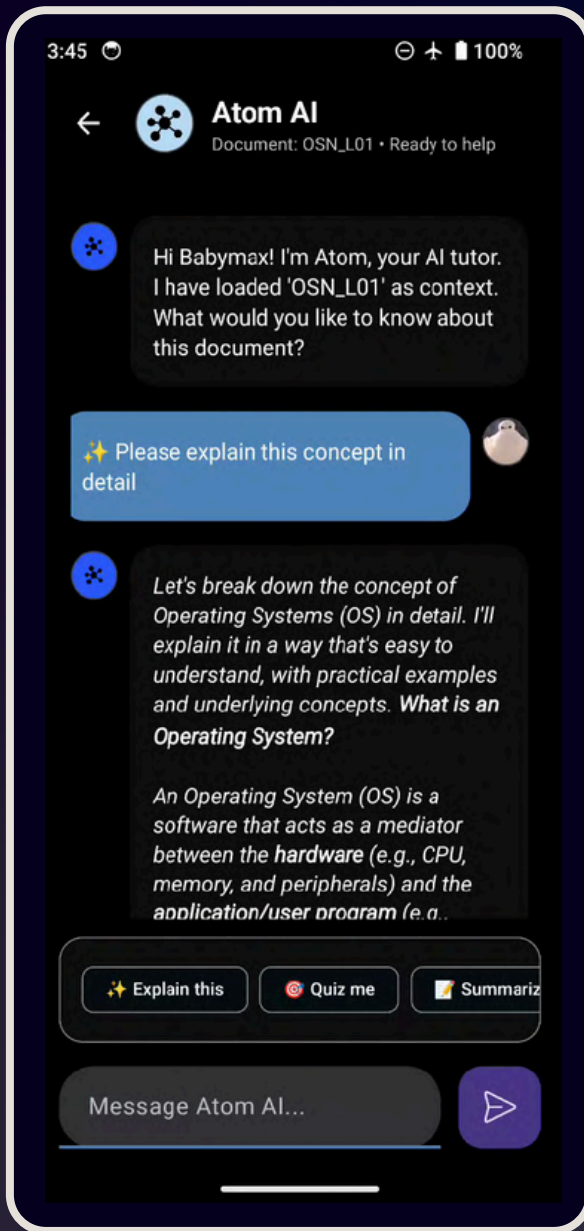
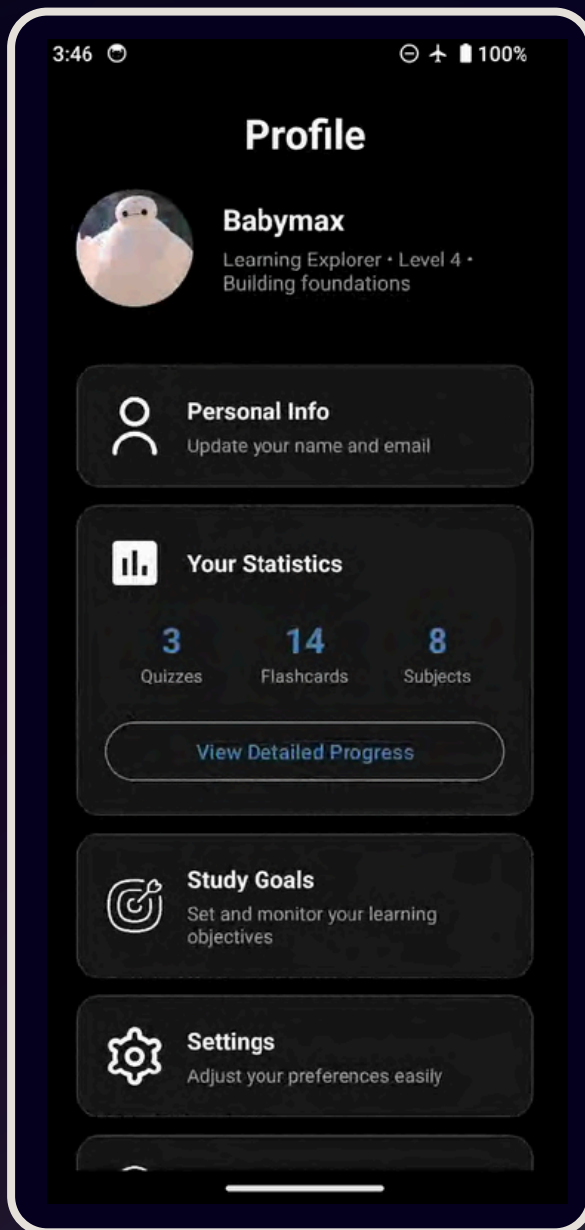
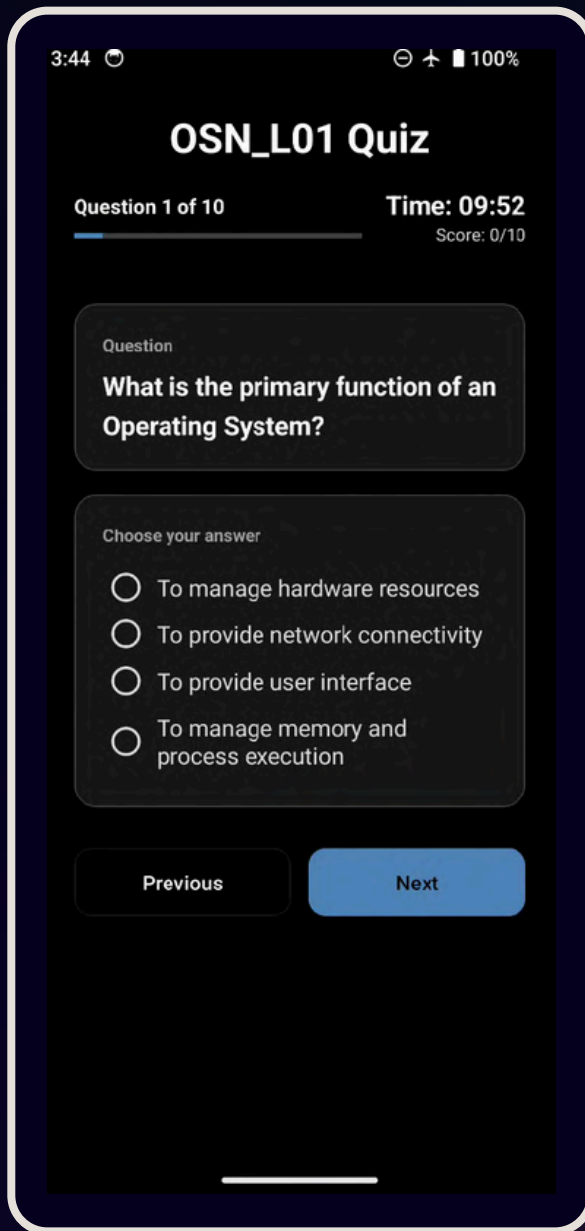
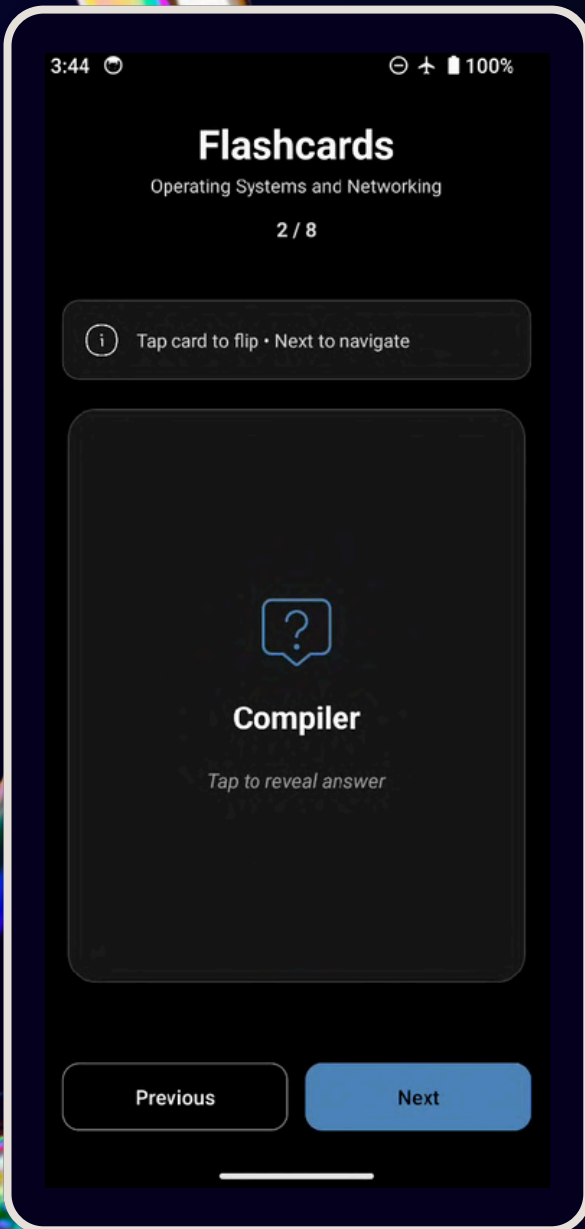
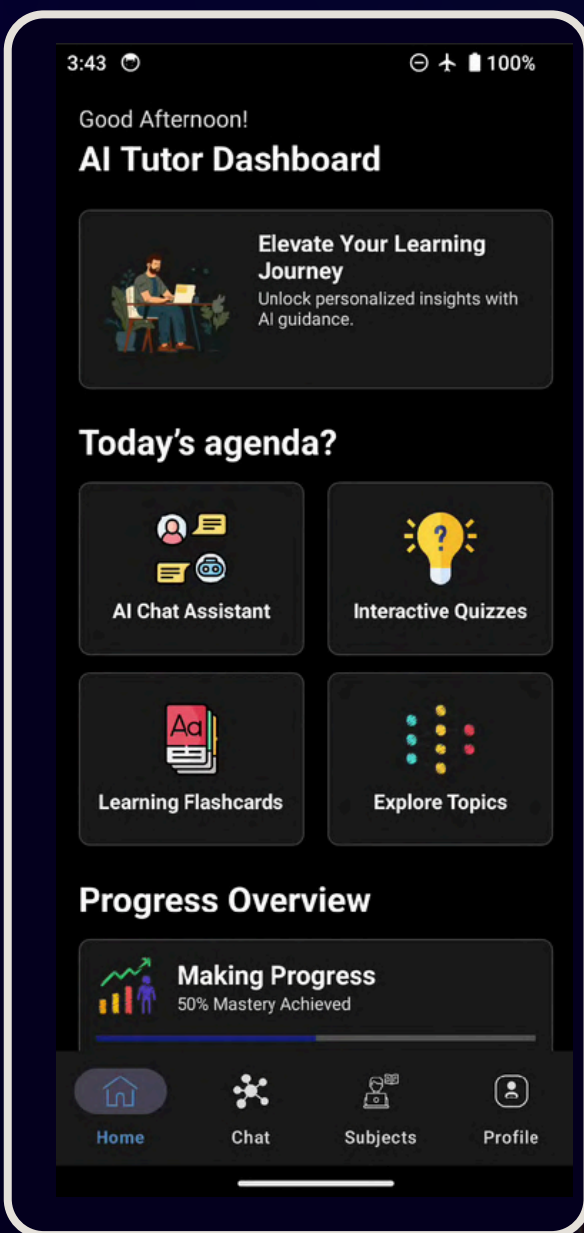
1 Quizzes	8 Flashcards	8 Subjects
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[View Detailed Progress](#)

Study Goals
Set and monitor your learning objectives

Settings
Adjust your preferences easily

APP FUNCTIONALITIES (Dark Mode)



Model Result

Metric	NPU (Genie)	CPU (llama.cpp)
Inference Speed (tokens / sec)	26-30	2-5
Time to First Token (TTFT) (ms)	45	200-500
Model Loading (sec)	2.7	5-15
Hardware	Qualcomm NPU	ARM64 CPU





THANK
YOU